gather outside the gates of under-utilised (or, even worse, closed) factories desperately seeking jobs.

At some point, the recession will be so deep that those firms which have survived will start doing rather well. The reason is that, with many of their competitors out of the market, they will be enjoying a much bigger share of the market. Even if the pie has shrunk there will be far fewer firms competing for pieces of it and therefore those still left in the game will get their hands on a larger piece of the (albeit shrinking) pie. Additionally, a recession reduces firms' cost by creating a large pool of idle capital and labour whose prices drop below its values. In plain language, during a recession surviving firms can purchase raw materials, computer equipment, machinery, etc. for a song. As for workers, desperate as they are for work they will labour for lower wages and, even if they receive the same wage as before, they will be working twice as hard fearing that, at the employer's whim, they may join the scrapheap of wasted humans who are knocking pathetically on the factory's gates. It is not at all surprising, in this theoretical context, that in Marx's eyes economic recessions are to capitalism what hell is to Christianty: indispensable.

However, unlike hell, an economic recession is not permanent. In an attempt to shore up their market dominance, surviving firms start expanding in the middle of the recession. As they expand (e.g. by employing more workers) they start another chain reaction, this time a positive one, which boosts output, employment and eventually capital accumulation. The economy thus exits the recession and enters a period of growth. But as before, this upturn contains the seeds of the next recession. And so on.

Marx went on to predict that every recession revitalises capitalism as it plays a culling role which helps the fitter companies survive at the expense of the fragile; a process that, many years after Marx, the Austrian economist Joseph Schumpeter (see p. 190) described as the process of 'creative destruction'. However, Marx believed that with every recession that passes the poverty and inequality left behind would worsen. And every new period of growth will be less likely to undo the social and economic damage caused by the previous recession. In time, capitalism will exhaust its capacity to innovate and to utilise resources (especially human resources) with any semblance of efficiency. The answer to this senseless cycle, Marx argues, is to devise a more rational economic system than capitalism: socialism. For this revolutionary thinker, capitalism is not 'wrong' because it is unfair; it is unfair because it is irrational and fundamentally wasteful.

1.2.5 The twin masters of economics: history and ideology

This is not a book on the history of economic thought. The point in skimming through the surface of the simultaneous development of capitalism and of economic ideas about capitalism is to give the reader a feeling for the precariousness of economic theory. Unlike physics, it is not simply a response to the need to know how an objective reality 'out there' works. Unlike nature which conveniently stays at an arm's length from our theories, society is less accommodating: our theories about society are so bound up with society itself that it is radically harder for a social scientist to create enough distance between her and the object of her study than it is for a natural scientist. The result is that economics is partly science and partly ideology. Which part is greater is also a source of contention amongst economists!

Can economics be a science?

Most economists answer with a resounding 'Yes'. They distinguish between two types of economics: *positive* and *normative*. Positive economics is proclaimed as the scientific study of how things are, of how particular economic systems work. Normative economics is about how we would like things to be, of which type of system we favour. (Textbooks make this distinction and then concentrate on positive economics. Indeed a famous textbook by Richard Lipsey is entitled *Positive Economics*.) The idea is that we keep positive economics free of ideology, ethical judgments, political passions, etc. and, once we have a clear picture of what is feasible, we can then let the passions in to decide what is desirable (that is, move on to normative economics).

Others (including me) disagree. They point out that the positive-normative distinction is impossible to hold on to. For example, they suggest that behind every piece of 'positive economic thinking' lies an ideological position. Even worse, and unlike in physics or chemistry, there can be no ideology-free economic facts. Take inflation, that is, the rate of increase in prices: which prices do we measure? Do we look at the price of a Rolls-Royce, of a bus ticket or a combination of the two? And if we choose the latter, how much emphasis should be given to the price of the Rolls relatively to bus fares when calculating the average change in the cost of transport? In other words, *every* measure of inflation hides a *political* decision as to which group of people matter more (e.g. the rich or the poor). And to try to pretend that economics is an objective study of the possible (as opposed to the desirable) is a political attempt to present certain politically loaded views as objective and thus superior.

Regardless of all this, economic textbooks seem strictly un-ideological. Yet every economic theory is based on some ideological or political prejudice. We saw how Adam Smith created a theory consistent with his commitment to free trade and the historical events of the eighteenth century which were marked by the rise of market societies out of feudal societies with markets. Smith takes the logic of the 1770s tenant (who rents land from the lord and hires wage labour), of the merchant, of the butcher and assumes that theirs is the logic of rational men (note that the rationality of women was hotly disputed on sexist philosophical grounds during that time) at all times and in all places. He forgets that this logic is also a product of history; that it did not exist until the tenant, the smith, the butcher and the worker were all forced by historical change to enter the competitive market and to develop a specifically market-oriented mentality. Adam Smith's *ideological* identification with this mentality spawned a particular view of human nature and of the type of social organisation (i.e. free market capitalism) which can serve it best. However impressed one is by Smith's vision, it is unwise to interpret it as an objective, scientific, unideological model of the social world.

David Ricardo's theory was also a product of his ideology and of the period during which he thought and wrote. It was utterly in tune not only with Ricardo's dislike for rentiers (i.e. landowners, like himself, who creamed off the benefits from capital accumulation without contributing anything to it) but also with the historical developments of his time (i.e. the Napoleonic Wars and the subsequent political tussle against protectionist landowners). Finally, Karl Marx's model of capitalism reflected not only his solidarity with the working class against its exploiters but also the history of recessions and revolutions during the period when he was writing.

Since then ideology and history have continued to spawn contradictory economic perspectives like those (to mention a few who probably mean nothing to you at the moment) of Rosa Luxemburg, Joseph Schumpeter, John Maynard Keynes, Friedrich von Hayek, Paul Sweezy, John Kenneth Galbraith, Joan Robinson, Milton Friedman and Robert Lucas. The reason why they disagree violently amongst them is not that some are more intelligent than others. Rather it is because their starting ideological position was different and because they did not have a shared history.

1.3 Modern textbook economics (or neoclassical economics)

1.3.1 The transition from classical to neoclassical economics

Imagine that you are the first ever Professor of Economics in some prestigious European university towards the end of the nineteenth century. After your first lecture you enter the Common Room where all the professors meet at tea-time. You sit around the table unobtrusively listening to the various conversations. The Professor of Physics is going on about the latest advances in thermodynamics and the exciting possibility of understanding the universe in a non-Newtonian manner. The Professor of Biology predicts that Darwinian theories will eventually focus on the evolution not of whole animals but of genes and perhaps of smaller entities which make up genes. Meanwhile the Professors of Philosophy, Law and Linguistics are immersed in witty exchanges on the nature of language. At some point, they notice your presence and interrupt their discussions. One of them shatters the awkward silence by asking: 'Tell us old boy, what is this economics science of yours? Is it worth the candle?'

What do you say? Do you give them a spiel on how when the demand for shoes exceeds the supply of shoes, shoes will appreciate in price? Or do you tell them about Adam Smith's escalator? Or David Ricardo's dislike of landowners? It is embarrassing, isn't it? I suspect that in that position you would have a great urge to convince these snobs that your discipline is as scientific and respectable as theirs. Regardless of whether the first academic economists actually felt that way, their theoretical endeavours are not incompatible with such feelings of embarrassment at what is now called classical political economy, or classical economics (i.e. the work of Adam Smith, David Ricardo, Karl Marx and others).

Although they recognised the gravity of the main economic ideas in the classical texts, they thought of them as too bound up with politics, ideology and guesswork; features which precluded the development of a discipline as well behaved and professional as, say, physics. In a short space of time, academic economists like Alfred Marshall (1842–1924) and Léon Walras (1834–1910) (see box on p. 32) recalibrated economic theory to fit into the mould of natural science. By the turn of the century economies' main new feature was the extensive use of mathematics and the explicit attempt to rid economics of the politics, the passion and the philosophical wanderings which people like Smith, Ricardo and Marx had woven into it. Today we refer to the product of their labours as neoclassical economics. Modern textbooks attempt to convey competently a synopsis of those *neoclassical* efforts from the end of the nineteenth century to date.

Imagine now that it was you who had the job of reworking economic theory so that it comes to resemble physics rather than politics and philosophy. How would you go about doing it? Borrowing ideas from physics on how to construct a distinctly 'scientific' approach would be a good start. At that time the dominant physics' model was that of classical mechanics as conceived by Isaac Newton (1642–1727). To cut a long and glorious story very short, Newtonian classical mechanics comprised four steps. Starting with a decision on what the focus of study ought to be (i.e. objects with certain physical characteristics like mass, location, velocity and so on), physics founded its great theoretical breakthrough on certain assumptions concerning the way nature worked (i.e. the Laws of Nature).

A founder of neoclassical economics speaks out

> As for those economists who do not know any mathematics... and yet have taken the stand that mathematics cannot possibly serve to elucidate economic principles, let them go their way repeating that 'human liberty will *never* allow itself to be cast into equations' or that 'mathematics ignores frictions which are *everywhere* in social science' and other equally forceful and flowery phrases. They can never prevent the theory of the determination of prices under free competition from becoming a mathematical theory. Hence, they will always have to face the alternative either of steering clear of this discipline and consequently elaborating a theory of applied economics without recourse to a theory of pure economics or of tackling the problems of pure economics without the necessary equipment, thus producing not only very bad pure economics but also very bad mathematics.
>
> Leon Walras, unpublished correspondence, 1900

And the first Professor of Economics at Cambridge offers a word of caution

> Most economic phenomena 'do not lend themselves easily to mathematical expression'. Economists must therefore guard against 'assigning wrong proportions to economic forces; those elements being most emphasised which lend themselves most easily to analytical methods'.
>
> Alfred Marshall, *Principles of Economics,* 1891

One of these assumptions is the *Principle of Energy Conservation* which, stated simply, suggests that energy is never born out of nothing and equally it never vanishes (this is why when a car crashes it explodes: kinetic energy does not evaporate but transforms itself into thermal energy). Notice how, at first, this is a theoretical proposition, an assumption. For all we know, it could be wrong. To find out more, physicists traced the repercussions of this assumption on the behaviour of the things it chose to study–of objects. They worked out mathematically what type of behaviour is compatible with the initial assumptions. For example they showed that for energy to be conserved the acceleration of an object subjected to some force must be a particular function of its mass and of the magnitude of that force. Finally, the time came where this whole model can be put to the test. If objects behaved in the laboratory according to the mathematical rules just derived, then the theory was to be accepted as true. Otherwise back to the drawing board. The question then becomes: how could we build an economic theory along these lines?

The structure of explanation in classical mechanics

Step 1 *Identify the focus of study*
Objects (e.g. atoms, molecules, electrons, a pendulum, etc.)
Step 2 *Articulate a grand theoretical assumption*
The *Principle of Energy Conservation* (i.e. energy neither vanishes into nothing nor is it born from nothing)
Step 3 *Describe mathematically the behaviour of our focus of study consistent with Step 2*
Mass=force times acceleration
(i.e. if a force is applied on some object, its acceleration will equal the ratio of its mass and the magnitude of the force)
Step 4 *Observe in the laboratory if actual objects behave according to Step 3*
If yes, accept the assumption in Step 2 and the theory in Step 3.

The structure of explanation in neoclassical economics

Step 1 *Identify the focus of study*
Decision makers (e.g. individuals, firms, organisations, governments, etc.)
Step 2 *Articulate a grand theoretical assumption*
The *Principle of Utility Maximisation* (i.e. decision makers strive to satisfy their preferences)
Step 3 *Describe mathematically the behaviour of our focus of study consistent with Step 2*
Marginal benefits=Marginal losses
(i.e. agents will do X until the last smidgen of X produces the same benefits as it does losses)
Step 4 *Use statistical methods (i.e. econometrics) to find out if Step 3 is correct*
If yes, accept the assumption in Step 2 and the theory in Step 3.